

Sumit Kumar

3rd Year Undergraduate

Department of Computer Science and Engineering

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Sumit Kumar SumitKumar-17

Academic Qualifications

Year	Degree/Certificate	Institute	CGPA/%
2022 - Present	5 year Dual Degree Course(BTech. + MTech.) in Computer Science	Indian Institute of Technology, Kharagpur	8.87/10
2022	Class XII (CBSE)	Delhi Public School, Kalyanpur	94.20%
2020	Class X (CBSE)	Delhi Public School, Kalyanpur	96.60%

Work Experiences

Spring Fest Website - Spring Fest

Aug'23-Jan'24

- Engineered a polished, **feature-rich website**, facilitating seamless **user authentication and login** for personalized festival experiences.
- Streamlined the **event registration process**, providing attendees with an intuitive and user-friendly interface for seamless event sign-up.
- Integrated** robust **secure payment gateways**, enabling **online transactions** for **accommodation bookings** and **merchandise purchases**.
- Established an interactive **online store** for **festival merchandise**, ensuring a seamless shopping experience with secure checkout.
- Implemented** a user **complaints submission** system to address and resolve attendee concerns, enhancing overall user satisfaction.

Key Projects

Raspberry Pi - Self project

May'24 - Jul'24

- Deployed four instances of website using **Docker** containers within **Kubernetes Pods** along with load balancing.
- Exposed** the cluster's target port from my **home network** to the wider **internet** via **port forwarding** on home router.
- Used Dynu's **Dynamic DNS** to dynamically map **non-static IP Address** of my home network with a **level 3 domain name**.
- Deployed a **DNS Server** using **Pihole** for content filtering which was used as DNS Server by **DHCP Server** in home router.

Online Classroom Management System - Course Project

Mar'24 - Apr'24

- Virtual classrooms feature a **messaging interface** for student queries, while professors can **upload lectures** and **assignments** exclusively.
- File-sharing system** ensures only professors can **upload content**, while students can **access files**. A personalized **calendar** displays schedules.
- Students can **download lectures**, **assignments**, and **tests**, and view **grades**. Option to **upload assignments** privately before deadlines.
- Incorporation of **role-based access control** ensures secure file handling and **data confidentiality** within the system.

Skills & Expertise

- Programming Languages:** C/C++, JavaScript, TypeScript, Python3, Bash, Java, Dart, Golang, Rust, Elixir, Awk
- Libraries and Frameworks:** ReactJS, NextJS, React Native, Flutter, NodeJs, Pandas, NumPy, C++ STL, GraphQL, Redis, MongoDB, SQL(Postgres, MySQL), Docker, Kubernetes
- Tools:** Git, GitHub, Linux Shell, Android Studio, Visual Studio Code, GDB, Valgrind, Jupyter

Relevant Courses

- Computer Science:** Programming and Data Structures, Algorithm-I, Algorithms Lab, Software Engineering, Software Engineering Laboratory, Computer Organization and Architecture
- Mathematics:** Advanced Calculus, Linear Algebra and Numerical and Complex Analysis, Probability & Statistics, Formal Language and Automata Theory
- Electronics:** Basic Electronics, Switching Circuits and Logic Design, Basic Electronics Laboratory, Switching Circuits Laboratory

Awards and Achievements

- Secured All India Rank of **2038** in Joint Entrance Examination Advanced, 2022 among 1.5 lakh students.
- Secured All India Rank of **1870** out of 1.2 million candidates in Joint Entrance Examination,Mains 2022.
- Secured All India Rank of **83** in WBJee, 2022 out of 1 lakh students.

Positions of Responsibility

Tech Head - Spring Fest

Aug'23 - till now

- Led website and app development for Spring Fest, IIT Kharagpur, implementing agile methodologies for efficient project management.
- Managed on-site technical operations during the festival, addressing real-time issues for a seamless digital experience for attendees
- Supervised event registration and coordinated attendee accommodations, showcasing effective leadership and organizational skills.

Extra Curricular Activities

- Member of the cricket team that won the Fresher's Tournament at IIT Kharagpur.
- Participated in General Championship, Table Tennis representing Vidyasagar Hall of Residence at IIT Kharagpur.
- Participated in General Championship, Stageplay, and Streetplay representing Vidyasagar Hall of Residence at IIT Kharagpur.
- Actively participated and contributed to Illumination 2023, the vibrant cultural pride of IIT Kharagpur.